

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 01
Duration: 25 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name: Solution	ID:	Section:06
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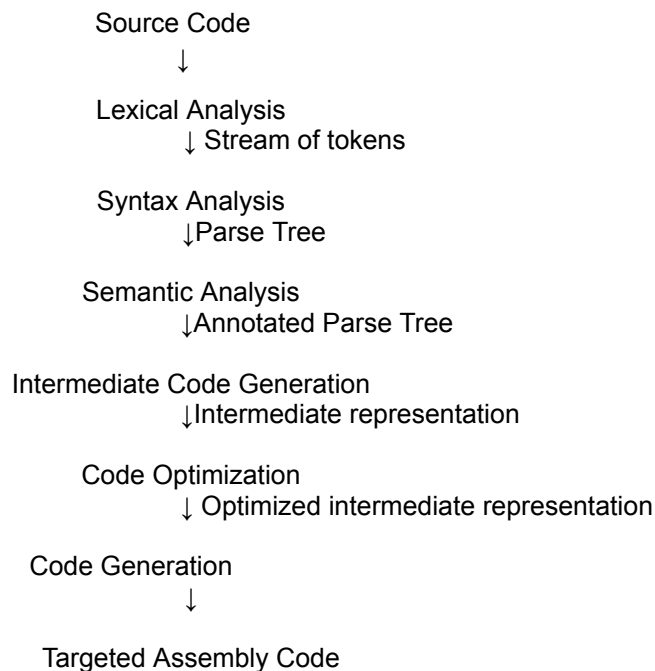
1. Define the analysis and synthesis model of compilation (2 x 3 = 6)
2. Draw the stages of a compiler. (4)
3. Find out how many lexemes are there in the below code? How many tokens will be generated for the below code? (5)

`int abc() { arr[10][3] = cd; }`

Question 01:

The analysis part breaks up the source program into constituent pieces and imposes a grammatical structure on them. It then uses this structure to create an intermediate representation of the source program. If the analysis part detects that the source program is either syntactically ill formed or semantically unsound, then it must provide informative messages, so the user can take corrective action. The analysis part also collects information about the source program and stores it in a data structure called a symbol table, which is passed along with the intermediate representation to the synthesis part. The synthesis part constructs the desired target program from the intermediate representation and the information in the symbol table.

Question 02:



Question 03:

No of lexemes: 16

No of tokens: 16